

MAT 528 HW09

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Due 11/13

Problem 1 (Munkres 33.3). Give a direct proof of the Urysohn lemma for a metric space (X, d) by setting

$$f(x) = \frac{d(x, A)}{d(x, A) + d(x, B)}.$$

Solution. Let A and B be disjoint closed sets and f as given. Recall that if C is a closed set, then $d(x, C) = 0$ is equivalent to $x \in C$, for $d(x, C) = 0$ implies $B_\varepsilon(x) \cap C$ is nonempty for all $\varepsilon > 0$, hence x is a limit point of C , so belongs to C (and the converse is trivial).

First, we show that f is continuous. Now observe that for any $C \subseteq X$ and $x, y \in X$,

$$d(x, C) = \inf_{z \in C} d(x, z) \leq \inf_{z \in C} \{d(x, y) + d(y, z)\} = d(x, y) + d(y, C)$$

and symmetrically $d(y, C) \leq d(x, y) + d(x, C)$, so $|d(x, C) - d(y, C)| \leq d(x, y)$. Thus, if $d(x, y) < \varepsilon$ then so is $|d(x, C) - d(y, C)|$, proving that $d(x, C)$ is continuous in x , meaning f is continuous at all $x \in X$ such that $d(x, A) + d(x, B)$ is nonzero. Since $d(x, A)$ and $d(x, B)$ are nonnegative, this can happen only if $d(x, A) = d(x, B) = 0$, i.e. $x \in A \cap B = \emptyset$, which is to say that $d(x, A) + d(x, B)$ is never zero. Therefore f is continuous on X .

To see that f separates A and B , we see that $x \in A$ implies $d(x, A) = 0$, so

$$f(x) = \frac{0}{0 + d(x, B)} = 0,$$

while $x \in B$ implies that $d(x, B) = 0$ and since $x \notin A$, $d(x, A) \neq 0$, hence

$$f(x) = \frac{d(x, A)}{d(x, A) + 0} = 1. \quad \square$$

Problem 2 (Munkres 36.1). Prove that every manifold is regular and hence metrizable. Where do you use the Hausdorff condition?

Solution. Let M be an m -manifold, $x \in M$, and $U \subseteq M$ a neighborhood of x . Then, there exists an open $W \subseteq M$ that is homeomorphic to open $V \subseteq \mathbb{R}^m$ via homeomorphism h ; take $W' = W \cap U$ and $V' = h(W')$ the homeomorphic image of W' . Since V' is open it contains a closed ball C centered at $h(x)$ (which, by Heine-Borel, is compact), and C contains an open ball B centered at $h(x)$. Since $h^{-1}|_{V'}: V' \rightarrow W'$ is continuous, $h^{-1}(C) \subseteq W'$ and $h^{-1}(B) \subseteq h^{-1}(C)$ are respectively compact and open, both containing x .

I claim that the closure of $h^{-1}(B)$ is contained in U : since $h^{-1}(C)$ is compact, it is closed in M , and by construction it contains $h^{-1}(B)$, so $h^{-1}(C)$ contains the minimal closed set (under inclusion) containing $h^{-1}(B)$, which by definition is the closure of $h^{-1}(B)$.

To summarize, we have shown that, given $x \in M$ and a neighborhood $U \subseteq M$ of x , there exists a neighborhood of x whose closure lies in U , which by Munkres's Lemma 31.1 is equivalent to M being a regular space. $\square$

Problem 3 (Munkres 43.5). Show that if f is a contraction of a complete metric space, then there is a unique point $x \in X$ such that $f(x) = x$.

Solution. First, we show that f has a fixed point. Fix $x_0 \in X$ and let $x_n = f(x_{n-1})$ for positive integer n . Since f is a contraction, there exists $\alpha \in (0, 1)$ such that $d(f(x), f(y)) \leq \alpha d(x, y)$ for all $x, y \in X$. If f^k is the function f composed with itself k times, then induction tells us $d(f^k(x), f^k(y)) \leq \alpha^k d(x, y)$. Letting $x_0 \in X$ be arbitrary and setting $x_n = f^n(x_0)$ for $n \in \mathbb{N}$, we get that for integers $m < n$,

$$d(x_n, x_m) \leq \sum_{j=m}^{n-1} d(x_j, x_{j+1}) \leq \sum_{j=m}^{n-1} \alpha^j d(x_0, x_1) \leq d(x_0, x_1) \cdot \sum_{j=m}^{\infty} \alpha^j = d(x_0, x_1) \cdot \frac{\alpha^m}{1 - \alpha}.$$

Since $\lim_{m \rightarrow \infty} \alpha^m = 0$, given $\varepsilon > 0$ there exists N such that $d(x_0, x_1) \cdot \alpha^N / (1 - \alpha) < \varepsilon$, so that $n > m > N$ implies

$$d(x_n, x_m) \leq d(x_0, x_1) \cdot \frac{\alpha^m}{1 - \alpha} \leq d(x_0, x_1) \cdot \frac{\alpha^N}{1 - \alpha} < \varepsilon.$$

This means (x_n) is a Cauchy sequence, so it converges to some $x_\infty \in X$. This x_∞ is a fixed point of f , since

$$x_\infty = \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} f(x_{n-1}) = f\left(\lim_{n \rightarrow \infty} x_{n-1}\right) = f(x_\infty)$$

where the limit can be pulled inside f because contractions are continuous (where $d(x, y) < \varepsilon$ implies $d(f(x), f(y)) \leq d(x, y) < \varepsilon$).

The fixed point is unique because if $x, x' \in X$ are distinct fixed points of f , then

$$d(x, x') \leq \alpha d(f(x), f(x')) = \alpha d(x, x') < d(x, x'),$$

which is impossible. □

Problem 4 (Munkres 43.7). Show that the set of all sequences $(x_1, x_2, \dots)$ such that $\sum_{k=1}^{\infty} x_k^2$ converges is complete in the ℓ^2 metric.

Solution. The set of all such sequences is a normed vector space with norm $\|\mathbf{x}\| = (\sum_{k=1}^{\infty} x_k^2)^{1/2}$, so $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ (as proven in HW03). Let $\mathbf{x}_n = (x_{n,1}, x_{n,2}, \dots)$, $n \in \mathbb{N}$, be a Cauchy sequence in ℓ^2 . Each of the sequences $(x_{m,k})_{m \in \mathbb{N}}$ is Cauchy because $|x_{m,k} - x_{n,k}| < \|\mathbf{x}_m - \mathbf{x}_n\| < \varepsilon$ for large enough m, n , so $\mathbf{x}_n$ converges coordinatewise to some $\mathbf{x} = (\ell_1, \ell_2, \dots)$ (i.e. $\ell_k = \lim_{m \rightarrow \infty} x_{m,k}$).

Claim 4.1 — The vector

$$\mathbf{x} = (\ell_1, \ell_2, \ell_3, \dots)$$

is in ℓ^2 .

Proof. We wish to show that

$$\sum_{i=1}^{\infty} |\ell_i|^2$$

converges. It suffices to show that the partial sums are bounded, because the numbers being summed are all nonnegative. For each $i \leq N$, there exists an M_i such that

$$n > M_i \implies |\ell_i - x_{n,i}| < 2^{-i}.$$

Fix $n > \max\{M_1, \dots, M_N\}$. The N th partial sum can be bounded above as follows:

$$\left(\sum_{i=1}^N |\ell_i|\right)^{1/2} \leq \left(\sum_{i=1}^N |\ell_i - x_{n,i}|\right)^{1/2} + \left(\sum_{i=1}^N |x_{n,i}|\right)^{1/2} < \left(\sum_{i=1}^{\infty} 2^{-i}\right)^{1/2} + \|\mathbf{x}_n\| = 1 + \|\mathbf{x}_n\|.$$

Additionally, Cauchy sequences are bounded, so $\|\mathbf{x}_n\|$ is bounded above by a constant C for all n . Therefore we have bounded the partial sums of the $|\ell_i|^2$ by $(1 + C)^2$, so the series converges. □

Now, we need to prove that $\mathbf{x}_n \rightarrow \mathbf{x}$ with respect to the ℓ^2 metric. Let $\varepsilon > 0$ and $N \in \mathbb{N}$. For each $i = 1, \dots, N$, we can find thresholds M_i for which

$$m > M_i \implies |\ell_i - x_{m,i}|^2 < \varepsilon^2 \cdot 2^{-i-2}.$$

This means that, if $m > \max\{M_1, \dots, M_N\}$, then

$$\left(\sum_{i=1}^N |\ell_i - x_{m,i}|^2 \right)^{1/2} < \left(\sum_{i=1}^{\infty} \varepsilon^2 \cdot 2^{-i-2} \right)^{1/2} = \sqrt{\varepsilon^2/4} = \varepsilon/2$$

We also know that $(\mathbf{x}_n)$ being Cauchy means that there is an M such that $m, n > M$ implies $\|\mathbf{x}_m - \mathbf{x}_n\| < \frac{\varepsilon}{2}$. Putting this together, we can fix m larger than M and all of the M_i and use the triangle inequality to say that $n > M$ implies

$$\begin{aligned} \left(\sum_{i=1}^N |\ell_i - x_{n,i}| \right)^{1/2} &\leq \left(\sum_{i=1}^N |\ell_i - x_{m,i}| \right)^{1/2} + \left(\sum_{i=1}^N |x_{m,i} - x_{n,i}| \right)^{1/2} \\ &\leq \varepsilon/2 + \|\mathbf{x}_m - \mathbf{x}_n\| < \varepsilon. \end{aligned}$$

In other words, as long as $n > M$ (noting that M only depends on ε), we can bound all the partial sums of N below ε , so $n > M$ implies that $\|\mathbf{x} - \mathbf{x}_n\| < \varepsilon$, which is what it means for $\mathbf{x}_n$ to converge to $\mathbf{x}$ in the metric of ℓ^1 , so we are done. $\square$

Problem 5 (Munkres 43.9). Let (X, d) be a metric space. Let $\tilde{X}$ denote the set of all Cauchy sequences $\mathbf{x} = (x_1, x_2, \dots)$ of points of X . Define $\mathbf{x} \sim \mathbf{y}$ if $d(x_n, y_n) \rightarrow 0$. Let $[\mathbf{x}]$ denote the equivalence class of $\mathbf{x}$, and let Y denote the set of these equivalence classes. Define a metric D on Y by the equation $D([\mathbf{x}], [\mathbf{y}]) = \lim_{n \rightarrow \infty} d(x_n, y_n)$.

- (a) Show that $\sim$ is an equivalence relation, and show that D is a well-defined metric.
- (b) Define $h: X \rightarrow Y$ by letting $h(x)$ be the equivalence class of the constant sequence $(x, x, \dots)$. Show that h is an isometric embedding.
- (c) Show that $h(X)$ is dense in Y ; indeed, given $\mathbf{x} = (x_1, x_2, \dots) \in \tilde{X}$, show that the sequence $h(x_n)$ of points of Y converges to the point $[\mathbf{x}]$ of Y .
- (d) Show that if A is a dense subset of a metric space (Z, ρ) and if every Cauchy sequence in A converges in Z , then Z is complete.
- (e) Show that (Y, D) is complete.

Solution. (a) Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \tilde{X}$. The relation is reflexive, since $\mathbf{x} \sim \mathbf{x}$ because $\lim_{n \rightarrow \infty} d(x_n, x_n) = \lim_{n \rightarrow \infty} 0 = 0$. The relation is symmetric because $\lim_{n \rightarrow \infty} d(x_n, y_n) = 0$ implies $\lim_{n \rightarrow \infty} d(y_n, x_n) = 0$ hence $\mathbf{y} \sim \mathbf{x}$. The relation is transitive because if $\mathbf{x} \sim \mathbf{y}$ and $\mathbf{y} \sim \mathbf{z}$ then

$$\lim_{n \rightarrow \infty} d(x_n, z_n) \leq \lim_{n \rightarrow \infty} d(x_n, y_n) + d(y_n, z_n) = \lim_{n \rightarrow \infty} d(x_n, y_n) + \lim_{n \rightarrow \infty} d(y_n, z_n) = 0,$$

so the limit on the left is zero, i.e. $\mathbf{x} \sim \mathbf{z}$.

To check that D is well-defined we need to check that if $\mathbf{x}_1 \sim \mathbf{x}_2$ and $\mathbf{y}_1 \sim \mathbf{y}_2$ then

$$\lim_{n \rightarrow \infty} d(x_{1,n}, y_{1,n}) = \lim_{n \rightarrow \infty} d(x_{2,n}, y_{2,n})$$

where $\mathbf{x}_j = (x_{j,1}, x_{j,2}, \dots)$ and $\mathbf{y}_j = (y_{j,1}, y_{j,2}, \dots)$ for $j = 1, 2$. By the triangle inequality,

$$\begin{aligned} d(x_{1,n}, y_{1,n}) &\leq d(x_{1,n}, x_{2,n}) + d(x_{2,n}, y_{2,n}) + d(y_{2,n}, y_{1,n}) \\ d(x_{2,n}, y_{2,n}) &\leq d(x_{2,n}, x_{1,n}) + d(x_{1,n}, y_{1,n}) + d(y_{1,n}, y_{2,n}), \end{aligned}$$

so we have the upper bound

$$|d(x_{1,n}, y_{1,n}) - d(x_{2,n}, y_{2,n})| \leq d(x_{1,n}, x_{2,n}) + d(y_{1,n}, y_{2,n}).$$

The right-hand side converges to zero because $\mathbf{x}_1 \sim \mathbf{x}_2$ and $\mathbf{y}_1 \sim \mathbf{y}_2$, hence so does the left-hand side, meaning $d(x_{1,n}, y_{1,n})$ and $d(x_{2,n}, y_{2,n})$ have the same limit.

(b) For any $x, y \in X$, $D(h(x), h(y)) = \lim_{n \rightarrow \infty} d(x, y) = d(x, y)$, so h is an isometric embedding.

(c) Since $\mathbf{x}$ is a Cauchy sequence, given $\varepsilon > 0$, there exists $N_\varepsilon \in \mathbb{N}$ such that $m, n > N_\varepsilon$ implies $d(x_m, x_n) < \varepsilon$. Then, if $n > N_\varepsilon$,

$$D(h(x_n), \mathbf{x}) = \lim_{k \rightarrow \infty} d(x_n, x_k) < \varepsilon$$

because the argument of the limit will be smaller than ε for all $k > N_\varepsilon$, which means $h(x_n)$ converges to $\mathbf{x}$. This implies $\mathbf{x}$ is a limit point of $h(X)$, so each point in Y is a limit point of $h(X)$, to elaborate on why $h(X)$ is dense in Y .

(d) Let (x_n) be a Cauchy sequence in Z and, for $\varepsilon > 0$, let $N_\varepsilon \in \mathbb{N}$ be such that $m, n > N_\varepsilon$ implies $\rho(x_m, x_n) < \varepsilon$. Since A is dense in Z , for each $n \in \mathbb{N}$, there exists $a_n \in A$ such that $\rho(a_n, x_n) < 1/n$ (since the open ball $B_{1/n}(x_n)$ must intersect A); since $\lim_{n \rightarrow \infty} \rho(a_n, x_n) = 0$, if either sequence (a_n) or (x_n) converges, then so does the other one and both sequences have the same limit. The sequence (a_n) is a Cauchy sequence: if $\varepsilon > 0$, pick $N \in \mathbb{N}$ such that $N > N_{\varepsilon/3}$ and $1/N < \varepsilon/3$, so $m, n > N$ implies

$$\rho(a_m, a_n) \leq \rho(a_m, x_m) + \rho(x_m, x_n) + \rho(x_n, a_n) < \frac{1}{m} + \frac{1}{n} + \frac{\varepsilon}{3} \leq \frac{2}{N} + \frac{\varepsilon}{3} < \varepsilon.$$

By assumption, then, (a_n) converges in Z , hence so does (x_n) , with $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} x_n$. Thus every Cauchy sequence (x_n) in Z has a limit in Z , so Z is complete.

(e) By (c), $h(X)$ is dense in Y and every Cauchy sequence $(h(x_n))$ in $h(X)$ converges to $[(x_n)] \in Y$, so, by (d) with $A = h(X)$ and $Z = Y$, Y is complete. $\square$

Problem 6 (Munkres 45.4a). Let $f_n: [0, 1] \rightarrow \mathbb{R}$ be the function $f_n(x) = x^n$. The collection $\mathcal{F} = \{f_n\}$ is pointwise bounded but the sequence (f_n) has no uniformly convergent subsequence; at what point or points does $\mathcal{F}$ fail to be equicontinuous?

Solution. $\mathcal{F}$ is equicontinuous at all $x_0 \in [0, 1)$ and not equicontinuous at $x_0 = 1$.

To show that $\mathcal{F}$ is (pointwise) equicontinuous on $[0, 1)$, we prove (pointwise) equicontinuity over $[0, c]$ for every $c \in [0, 1)$. Fix $c \in [0, 1)$. For each $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $c^N < \varepsilon$, hence $|x^n - 0| < \varepsilon$ for all $n > N$ and $x \in [0, c]$; this means that x^n converges to the zero function uniformly over $[0, c]$. The zero function is the only nontrivial limit point $\mathcal{F}|_{[0, c]} := \{f_n|_{[0, c]} : n \in \mathbb{N}\}$, since any limit point would be a limit of a subsequence of the $f_n|_{[0, c]}$, which is equal to the limit of the $f_n|_{[0, c]}$ (i.e. the zero function). That is to say, the closure of $\mathcal{F}|_{[0, c]}$ in the space of continuous functions over $[0, c]$ consists of $\mathcal{F}|_{[0, c]}$ together with the zero function.

This closure is easily seen to be compact, as, given an open cover of $\overline{\mathcal{F}}|_{[0, c]}$, whichever open set contains the zero function must also contain all but finitely many elements of the closure; and then the other elements can each have their own open set reserved for them to obtain a finite subcover. Now, Ascoli's theorem (Munkres's Theorem 45.4) tells us directly that $\mathcal{F}|_{[0, c]}$ is equicontinuous, meaning $\mathcal{F}$ is equicontinuous on $[0, c)$ for every $c \in [0, 1)$, hence equicontinuous on $[0, 1)$.

Now, to show that $\mathcal{F}$ is not equicontinuous at 1, suppose for contradiction that there exists U open in $[0, 1]$ and containing 1 such that $1 - x^n < \frac{1}{2}$ holds for all $x \in U$ and $n \in \mathbb{N}$. However, if one picks $x \in U$ not equal to (hence smaller than) 1, then $\lim_{n \rightarrow \infty} x^n = 0$, so for each $x \in U \setminus \{1\}$ there exists n such that $x^n < \frac{1}{2}$, i.e. $1 - x^n > \frac{1}{2}$; this proves that $\mathcal{F}$ is not equicontinuous at 1. $\square$